Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Development of an Integrated System Incorporating a Field-Adaptive AI Agent and High-Precision Automated Equipment for 3D Electroplating Quality Management**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Services		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	R&D Engineering Services	Product Quality Management Technologies	Chemical Processes	Process Equipment Technologies
Concept of Proposed Technology	▪ **Core Concept** - This project aims to develop an integrated electroplating equipment system in which a field-adaptive AI agent is natively embedded, through the simultaneous development of high-precision automated equipment and AI functions for 3D electroplating quality management in real manufacturing environments. - Unlike conventional approaches that apply AI as an external inspection or post-processing tool, the proposed AI agent is co-designed and co-implemented with the electroplating equipment, enabling continuous learning from field data and direct execution of adaptive quality control actions at the equipment level			

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▪ **Main function**

- Integrated acquisition and processing of high-resolution 3D electroplating quality data through sensing and measurement functions embedded within high-precision electroplating equipment
- Field-adaptive quality assessment performed by an AI agent that is directly integrated into the equipment control architecture and continuously updated using on-site operational data
- AI-driven decision-making for electroplating quality management that is jointly designed with equipment control logic, enabling adaptive determination of quality criteria and control strategies
- Direct execution of AI-based quality control actions through built-in automated equipment functions, ensuring stable and consistent electroplating quality without reliance on external systems
- Closed-loop operation in which 3D measurement, AI-based analysis, equipment-level control, and feedback learning are tightly coupled within a single integrated electroplating system

▪ **Expected uses of technology**

- Development and deployment of electroplating equipment in which field-adaptive AI functions are natively embedded within the hardware, enabling real-time 3D quality management and adaptive control directly at the equipment level rather than as an external or add-on software solution
- Application of the developed technology to existing and newly manufactured electroplating equipment, allowing on-site adaptation of quality management functions through tight integration between mechanical systems, control units, and AI-based decision logic
- Utilization by manufacturers with practical experience in electroplating equipment design, fabrication, and on-site commissioning, ensuring that the AI agent is co-developed and validated together with the physical equipment under real operating conditions
- Use of the integrated AI-enabled equipment to enhance production stability, reduce quality variation caused by environmental and process fluctuations, and minimize reliance on operator-dependent inspection and tuning
- Establishment of a scalable foundation for next-generation intelligent electroplating equipment, where field-adaptive AI capabilities are continuously updated and optimized through accumulated operational data across deployed systems

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Rationale for Joint R&D Project	▪ The development of a field-adaptive AI-enabled electroplating equipment system requires the close integration of advanced AI methodologies and high-precision equipment engineering, which necessitates an international joint R&D framework that combines complementary expertise across institutions. ▪ While the electroplating equipment manufacturer leads to the design, fabrication, and on-site validation of high-precision automated equipment, the international research partner contributes advanced AI agent design methodologies, adaptive learning algorithms, and data-driven quality management frameworks that are difficult to develop independently within a single organization. ▪ Electroplating quality management is highly dependent on equipment-specific characteristics and real-world operating conditions. Therefore, AI agents must be co-developed and validated using real manufacturing data and physical equipment, requiring continuous collaboration between the domestic equipment developer and the international AI research partner. ▪ Conventional approaches based on sequential development or externally sourced AI solutions face limitations in adaptability and scalability. Through international joint R&D, this project enables iterative feedback between equipment operation and AI model advancement, leveraging global research expertise while ensuring practical field applicability. ▪ The proposed international joint R&D structure is designed to establish a globally competitive intelligent electroplating equipment platform, embedding field-adaptive AI capabilities as core system components and facilitating future international technology diffusion and collaboration.
Key Words	1. Field-adaptive AI agent 2. AI-enabled electroplating equipment system 3. 3D electroplating quality management 4. High-precision automated equipment 5. Defect detection of 3D electroplating

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)
R&D Objectives	▪ To establish a systematic framework for collecting, preprocessing, and database construction of diverse 3D electroplating quality data generated directly from high-precision electroplating equipment operating under real manufacturing conditions ▪ To develop an integrated electroplating equipment system in which a field-adaptive AI agent is natively embedded, enabling real-time 3D electroplating quality management under actual manufacturing conditions ▪ To simultaneously design and implement high-precision automated electroplating equipment and AI-based quality management functions, ensuring tight coupling between equipment hardware, control architecture, and AI decision logic ▪ To establish a field-adaptive AI framework that continuously learns from on-site operational and quality data, allowing the equipment to maintain consistent plating quality across varying products, processes, and environmental conditions ▪ To validate the developed system through on-site deployment and operational testing, demonstrating its effectiveness in reducing quality variation, operator dependency, and manual inspection effort ▪ To create a scalable foundation for next-generation intelligent electroplating equipment, facilitating future expansion to additional quality attributes and process types			
R&D Target Specifications	▪ Development of a high-precision electroplating equipment platform capable of natively generating high-resolution 3D electroplating quality data, including geometric features, surface conditions, and thickness-related information, during normal equipment operation ▪ Establishment of a standardized pipeline for 3D electroplating data collection, preprocessing, refinement, and database construction, ensuring data consistency, traceability, and usability for AI learning and performance evaluation ▪ Implementation of a field-adaptive AI agent embedded within the equipment control architecture, capable of learning from accumulated 3D quality and process data and performing real-time quality assessment and anomaly detection ▪ Integration of AI-driven decision logic with equipment-level automation and control functions, enabling direct execution of adaptive quality control actions without reliance on external inspection or analysis systems ▪ Demonstration of robust and stable system performance under varying products, process parameters, and environmental conditions through on-site pilot operation and validation on real electroplating production lines ▪ Provision of an expandable and scalable system architecture that supports future			

	enhancement of AI models, data volume, and quality management functions, as well as potential application to additional surface treatment processes
R&D Plan	▪ **Core Technology Foundation (1-2 Years)** - Design and development of high-precision electroplating equipment architecture that supports native integration of 3D sensing, data acquisition, and AI-enabled control functions - Development of 3D electroplating quality data acquisition modules and establishment of initial data collection pipelines under real manufacturing conditions - Definition and implementation of data preprocessing, refinement, and database construction processes for systematic management of diverse 3D electroplating quality data - Initial development of a field-adaptive AI agent framework, focusing on 3D quality assessment, anomaly detection, and learning from on-site data - Integration of basic AI functions with equipment-level control architecture and validation through laboratory-scale and pilot equipment testing ▪ **Integrated System Development and Filed Validation (3-4 Years)** - Full-scale integration of the field-adaptive AI agent into the electroplating equipment control system, enabling real-time quality assessment and AI-driven decision-making - Expansion and refinement of the 3D electroplating quality database through continuous data accumulation from pilot and on-site operations - Enhancement of AI agent capabilities for adaptive quality decision-making and control strategy optimization under varying products, process parameters, and environmental conditions - Deployment of the integrated system on actual electroplating production lines and execution of on-site validation and performance evaluation - Collaborative refinement of AI models and control logic through international joint R&D activities, incorporating feedback from real equipment operation data ▪ **System Optimization, Scalability, and Commercial Readiness (5 Year)** - Optimization of the integrated AI-enabled electroplating equipment system for stability, robustness, and long-term operation in real manufacturing environments - Finalization of standardized 3D electroplating quality data management and AI operation frameworks suitable for commercial deployment - Demonstration of quantified improvements in quality consistency, defect reduction, and operational efficiency through extended field operation - Preparation of commercial-grade intelligent electroplating equipment, including

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	system documentation, operational guidelines, and deployment strategies - Establishment of a scalable and extensible platform for future application to additional electroplating processes, surface treatment technologies, and international markets

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	▪ Develop and provide advanced AI agent methodologies, including adaptive learning algorithms, decision-making frameworks, and data-driven quality management models suitable for field deployment ▪ Collaborate on the design of AI architectures and learning strategies that can be directly embedded into electroplating equipment control systems ▪ Support the analysis and utilization of accumulated 3D electroplating quality data to improve AI learning performance, robustness, and adaptability across varying operating conditions ▪ Participate in joint validation and performance analysis of AI-enabled quality management functions using real equipment operation data ▪ Contribute to knowledge exchange, joint publications, and international collaboration activities, enhancing the global competitiveness and technical credibility of the developed system
Expected R&R of Korean SME Partner	▪ Lead the overall system design and development of the integrated electroplating equipment, including mechanical structure, automation systems, control architecture, and on-site deployment ▪ Design, fabricate, and commission high-precision electroplating equipment capable of native integration with 3D sensing, data acquisition, and AI-enabled control functions ▪ Establish and operate the 3D electroplating quality data generation pipeline, including real-time data collection, preprocessing, refinement, and database construction during actual equipment operation ▪ Integrate the field-adaptive AI agent into the equipment-level control and automation system, ensuring real-time execution of AI-driven quality management and control actions ▪ Conduct on-site pilot operation, system validation, and performance evaluation under real manufacturing conditions, and lead activities related to system optimization and commercialization readiness

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	▪ **Scientific and Technological Effects** - Advancement of field-adaptive AI agent technologies through their native integration into high-precision electroplating equipment, moving beyond conventional rule-based or externally attached AI solutions - Establishment of a systematic 3D electroplating quality data lifecycle, including data generation, preprocessing, database construction, and continuous learning, enabling sustainable AI-driven quality management - Demonstration of a closed-loop integration framework that tightly couples 3D measurement, AI-based decision-making, and equipment-level control within a single manufacturing system - Enhancement of the robustness and adaptability of AI models by validating their performance under real manufacturing environments with diverse products, processes, and operating conditions ▪ **Economic and Industrial Effects** - Strengthening the global competitiveness of domestic electroplating equipment manufacturers by embedding AI capabilities directly into core equipment products, creating high value-added intelligent manufacturing equipment - Reduction of quality-related costs, including defect rates, rework, and operator-dependent inspection effort, leading to improved productivity and operational efficiency in electroplating processes - Acceleration of AI adoption among small and medium-sized enterprises (SMEs) by providing equipment-level AI solutions that do not require extensive in-house AI expertise or system integration capabilities - Creation of new business and commercialization opportunities for AI-enabled electroplating equipment in domestic and international markets, supporting export expansion and industrial upgrading ▪ **Policy and Social Effects** - Contribution to national policies promoting smart manufacturing and AI-driven industrial innovation, particularly through the integration of AI technologies into traditional manufacturing equipment - Support for the digital transformation of manufacturing SMEs, addressing challenges related to workforce shortages, skill gaps, and quality consistency - Enhancement of industrial safety and working conditions by reducing reliance on manual inspection and repetitive quality management tasks

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	Promotion of international research collaboration and technology exchange, reinforcing Korea's role in global intelligent manufacturing ecosystems